

FEATURE LEARNING, EFFECTIVE DIMENSION, AND ARCHITECTURE-SPECIFIC CRITICAL DEPTH IN TWO-LAYER RELU NETWORKS

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ABSTRACT. The minimax rate for nonparametric regression of C^s functions on $\mathbb{R}^d$ is $n^{-2s/(2s+d)}$, which deteriorates rapidly with the ambient dimension. Practical neural networks trained on high-dimensional inputs nonetheless generalize at rates that do not exhibit such severe dimension dependence. We give a precise mechanistic account of this gap for the *single-index* target class $f^*(\mathbf{x}) = \rho(\langle \mathbf{v}^*, \mathbf{x} \rangle)$ under Gaussian inputs, working within the noise-propagation framework of [1]. Our principal contribution (Theorem 3.1) shows that, conditional on the post-alignment hypothesis established in [2], the test risk of two-layer ReLU networks transitions from the kernel rate $\Theta(n^{-2s/(2s+d)})$ in the lazy regime to $\Theta(n^{-2s/(2s+1)})$ in the feature-learning regime; the proof routes the kernel rate through spherical-harmonic multiplicities $N_l \asymp l^{d-2}$ and the feature-learning rate through the rank-one effective dimension induced by the collapsed equivalent kernel. We then introduce an architecture-specific critical depth L^* (Theorem 6.2) defined as the smallest depth at which the effective rank of the noise propagation operator falls below $n^{2s/(2s+1)}$, and we derive $L_{\text{FC}}^* = \lceil (d-1)/2 \rceil$ for fully-connected ReLU networks, a depth-decoupled formula for residual networks, and $L_{\text{Transformer}}^* \in [1, 4]$ for $d \leq 512$. Finally, Theorem 10.2 extends the entire analysis to the cross-entropy loss via a Fisher-corrected operator $\mathbf{M}_{\text{CE}}$ and shows that, at the Neural Collapse fixed point, the effective rank equals $K-1$ for K -class classification, recovering dimension-free generalization without any further assumption.

1. INTRODUCTION

1.1. The dimension dependence of the kernel rate. A long-standing tension in the theory of overparameterized learning concerns the role

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of the ambient input dimension d . Classical nonparametric estimation gives a sharp lower bound: for the Hölder class $C^s(\mathbb{R}^d)$, no estimator can attain a rate better than $n^{-2s/(2s+d)}$, and kernel ridge regression with a smooth rotation-invariant kernel attains exactly this rate. The exponent is dominated by d for any fixed smoothness s , so that even modest inputs (e.g. $d = 100$) require an exponentially large number of samples for moderate accuracy. This is the *curse of dimensionality*.

Empirical practice presents a different picture. Networks trained on inputs of dimension d in the hundreds or thousands typically attain test errors that scale far better than $n^{-2s/(2s+d)}$. Two complementary explanations are usually offered. The first is structural: real targets are well approximated by low-complexity composites (single-index, multi-index, or sparse functions of low intrinsic dimension), so that the relevant rate is governed by an intrinsic dimension $k \ll d$. The second is algorithmic: gradient-based training does not produce a fixed-kernel estimator; rather, it performs feature learning, and the post-training Jacobian concentrates spectral mass on directions that are aligned with the target structure.

The first explanation is essentially structural and rests on assumptions about the data-generating process. The second, however, makes a more delicate prediction: even when the input is genuinely d -dimensional, an overparameterized network can attain a rate that scales as if the input were one-dimensional, provided that feature learning succeeds at aligning the weights with the target direction. The present paper makes this prediction quantitative and unconditional, given the alignment guarantees already established in [2].

1.2. Setting and prior work. We work within the framework introduced in [1], which models the test mean squared error of an interpolating predictor through the *noise propagation operator* $\mathbf{M} = \mathbf{\Sigma}^{-1}\mathbf{\Gamma}\mathbf{\Sigma}^{-1}$, where $\mathbf{\Sigma}$ is the diagonal matrix of singular values of the training Jacobian and $\mathbf{\Gamma}$ is the test–train cross-correlation matrix of gradient features. The trace $\text{tr}(\mathbf{M})$ governs the noise contribution to test error, and the benign overfitting trichotomy ($\beta < \alpha + 1/2$ catastrophic, $\beta = \alpha + 1/2$ tempered, $\beta > \alpha + 1/2$ benign) follows from the power-law decay of singular values $\sigma_j \asymp j^{-\alpha}$ and gradient-feature norms $\rho_j \asymp j^{-\beta}$. The companion paper [2] establishes the alignment hypothesis: in the rich regime, gradient flow drives the weights of every active neuron of a two-layer ReLU network into the span of the teacher direction $\mathbf{v}^*$.

We refer to [1] for the unified generalization bound and the benign/tempered/catastrophic dichotomy, and to [2] for the alignment lemma and its proof via ReLU directional independence. The present

paper does *not* reprove these results: we cite them as black-box inputs and develop their consequences.

The literature on dimension-free rates for neural networks is now substantial. Bach [4] analyzed mean-field two-layer networks and derived dimension-adaptive approximation rates. Bietti and Bruna [7] studied spherical harmonic decompositions of the NTK and gave matching upper and lower bounds on kernel ridge regression. Damian, Lee, and Soltanolkotabi [9] proved that one-pass SGD learns single-index targets at a rate determined by the information exponent. Ben Arous, Gheisari, and Jagannath [6] characterized the sample complexity of single-index recovery in the presence of high-dimensional noise. Rakhlin and Zhai [17] showed that interpolation by Laplace kernels is consistent only in high dimension, isolating the dimension dependence of benign overfitting in fixed-kernel regimes. Pappas, Han, and Donoho [15] introduced the Neural Collapse phenomenon, in which last-layer features and class means collapse onto a simplex equiangular tight frame at the terminal phase of training. Lu and Steinerberger [13] and others have given mathematical characterizations of the Neural Collapse fixed point.

1.3. Contributions. This paper makes three contributions, all conditional on the alignment hypothesis of [2].

- (i) **Feature-learning dimension collapse** (Theorem 3.1). For single-index targets with smoothness s , we show that the test risk transitions from the kernel rate $\Theta(n^{-2s/(2s+d)})$ to the dimension-free rate $\Theta(n^{-2s/(2s+1)})$ when feature learning succeeds. The proof identifies the spherical-harmonic multiplicity $N_l \asymp l^{d-2}$ as the source of the curse of dimensionality in the kernel regime, and the multiplicity collapse to a one-dimensional kernel as the mechanism responsible for the dimension-free rate after alignment.
- (ii) **Architecture-specific critical depth** (Theorem 6.2). We define L_A^* as the smallest depth at which the effective rank $r_{\text{eff}}(\mathbf{M}_A)$ falls below $n^{2s/(2s+1)}$ and derive separate formulae for three families: $L_{\text{FC}}^* = \lceil (d-1)/2 \rceil$ for fully-connected ReLU stacks (via depth-multiplicative spectral decay), L_{ResNet}^* decoupled from depth (via the identity-bypass structure of skip connections), and $L_{\text{Transformer}}^* \in [1, 4]$ for input dimension $d \leq 512$ (via the softmax-attention contribution to the cross-correlation spectrum). Each architecture requires a distinct proof structure.

(iii) **Cross-entropy extension via Neural Collapse** (Theorem 10.2).

We define a Fisher-corrected noise propagation operator $\mathbf{M}_{\text{CE}}$ and prove that the benign/tempered/catastrophic trichotomy of [1] is preserved verbatim under cross-entropy loss. Specializing to the Neural Collapse fixed point, we compute the effective rank $r_{\text{eff}}(\mathbf{M}_{\text{CE}}) = K - 1$ for K -class classification.

The three contributions together describe the boundary of feature learning: when alignment occurs, generalization is dimension-free; when it occurs at the Neural Collapse fixed point, the residual rank is exactly the number of classes minus one. The architectural results show that the depth required to enter the benign regime depends sensitively on the architecture, and that residual and attention mechanisms reduce this depth dramatically relative to plain stacks.

1.4. Organization. Section 2 recalls the noise propagation framework of [1] and the alignment hypothesis of [2]. Section 3 states Theorem 3.1, the feature-learning dimension collapse. Sections 4 and 5 prove the kernel and feature-learning halves of Theorem 3.1 respectively. Section 6 states Theorem 6.2; Sections 7, 8, and 9 prove the FC, ResNet, and Transformer cases. Section 10 states and proves Theorem 10.2 for the cross-entropy extension. Section 11 discusses limitations and open directions.

2. PRELIMINARIES

2.1. Recap of the noise propagation framework. We follow the notation of [1]. Let $f(\cdot; \theta): \mathbb{R}^d \rightarrow \mathbb{R}$ be a parametric predictor with parameters $\theta \in \mathbb{R}^p$, and let $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ be a training set with $y_i = g^*(\mathbf{x}_i) + \xi_i$, ξ_i i.i.d. sub-Gaussian with variance proxy σ^2 . Let θ^* be a minimum-norm interpolating parameter and let

$$\mathbf{J} \in \mathbb{R}^{n \times p}, \quad \mathbf{J}_{i,:} = \nabla_{\theta} f(\mathbf{x}_i; \theta^*)^{\top},$$

be the training Jacobian, with SVD $\mathbf{J} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\top}$. The j -th gradient feature is $\psi_j(\mathbf{x}) = \nabla_{\theta} f(\mathbf{x}; \theta^*)^{\top} \mathbf{v}_j$, the cross-correlation matrix is $\mathbf{\Gamma}_{jl} = \mathbb{E}_{\mathbf{x}}[\psi_j(\mathbf{x})\psi_l(\mathbf{x})]$, and the noise propagation operator is

$$\mathbf{M} = \mathbf{\Sigma}^{-1} \mathbf{\Gamma} \mathbf{\Sigma}^{-1} \in \mathbb{R}^{n \times n}.$$

Theorem 1 of [1] states that for any $\delta \in (0, 1)$, with probability $1 - \delta$,

$$(1) \quad \text{MSE}_{\text{test}} \leq C(B_{\text{signal}}^2 + \sigma^2 \text{tr}(\mathbf{M}) + \sigma^2 \|\mathbf{M}\|_F \sqrt{\log(1/\delta)} + \delta_{\text{lin}}^2).$$

Theorem 2 of [1] states that under power-law decay $\sigma_j \asymp j^{-\alpha}$, $\rho_j \asymp j^{-\beta}$, with $\rho_j = \sqrt{\mathbf{\Gamma}_{jj}}$,

$$\text{tr}(\mathbf{M}) < \infty \iff \beta > \alpha + \frac{1}{2}.$$

Definition 2.1 (Effective rank). For a positive semidefinite operator $\mathbf{A}$ with non-negative eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$, the *effective rank* is

$$r_{\text{eff}}(\mathbf{A}) := \frac{(\sum_j \lambda_j)^2}{\sum_j \lambda_j^2} = \frac{(\text{tr } \mathbf{A})^2}{\|\mathbf{A}\|_F^2}.$$

For $\mathbf{M}$, the effective rank quantifies the number of significant noise-propagation directions; benign overfitting requires $r_{\text{eff}}(\mathbf{M})$ to be at most polylogarithmic in n .

2.2. Single-index targets and the alignment hypothesis.

Definition 2.2 (Single-index target). A function $g^*: \mathbb{R}^d \rightarrow \mathbb{R}$ is a *single-index target* with link ρ and direction $\mathbf{v}^*$ if there exist $\mathbf{v}^* \in \mathbb{S}^{d-1}$ and $\rho: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$g^*(\mathbf{x}) = \rho(\langle \mathbf{v}^*, \mathbf{x} \rangle) \quad \text{for all } \mathbf{x} \in \mathbb{R}^d.$$

We say $g^* \in \mathcal{H}^s$ if $\rho \in C^s(\mathbb{R})$ with bounded derivatives up to order s .

We work throughout under the standard Gaussian input distribution $\mathbf{x} \sim \mathcal{N}(0, \mathbf{I}_d)$, with link function ρ that is non-affine and has information exponent $\kappa = 1$ in the sense of [6]; that is, the first non-trivial Hermite coefficient of ρ is non-zero. This is the regime in which the alignment hypothesis of [2] applies.

Assumption 2.3 (Alignment, after [2]). Let $f_\theta(\mathbf{x}) = p^{-1/2} \sum_{j=1}^p a_j \text{ReLU}(\mathbf{w}_j^\top \mathbf{x})$ be a two-layer ReLU student. Suppose the inputs are $\mathbf{x} \sim \mathcal{N}(0, \mathbf{I}_d)$ and the target is single-index $g^*(\mathbf{x}) = \rho(\langle \mathbf{v}^*, \mathbf{x} \rangle)$ with $\kappa = 1$. Suppose f_θ is trained by population gradient flow from a small-scale initialization. Then at any limit point θ^* of the gradient flow, the active neurons satisfy $\mathbf{w}_j \in \text{span}(\mathbf{v}^*)$, and the dead neurons satisfy $\|\mathbf{w}_j\| = O(\alpha_{\text{init}})$.

This is exactly the conclusion of Theorem 5a–c of [2]; we use it as a black box. The point of Theorem 3.1 below is to quantify the consequence of Assumption 2.3 for the test risk.

2.3. Equivalent kernels in the lazy and feature-learning regimes.

Definition 2.4 (Equivalent kernel). The *equivalent kernel* of an interpolating predictor at θ^* is the population kernel

$$K_{\theta^*}(\mathbf{x}, \mathbf{x}') := \mathbb{E}_{\theta^*} [\nabla_\theta f(\mathbf{x}; \theta^*)^\top \nabla_\theta f(\mathbf{x}'; \theta^*)],$$

where the expectation is taken over the randomness in the trained parameters (e.g. over initializations or training noise). When f_θ is in the lazy (NTK) regime, $K_{\theta^*} \approx K_{\text{NTK}}$ is the rotation-invariant Neural Tangent Kernel of [12]. After feature learning, K_{θ^*} is no longer rotation-invariant.

Lemma 2.5 (Spherical harmonic decomposition of the NTK; [7]). *Let K_{NTK} be the NTK of a two-layer ReLU network with input dimension d and Gaussian inputs. Then K_{NTK} admits the spherical-harmonic eigendecomposition*

$$K_{\text{NTK}}(\mathbf{x}, \mathbf{x}') = \sum_{l=0}^{\infty} \lambda_l^{(d)} \sum_{m=1}^{N(d,l)} Y_{l,m}(\mathbf{x}) Y_{l,m}(\mathbf{x}'),$$

where $\{Y_{l,m}\}$ is the orthonormal basis of spherical harmonics of degree l on $\mathbb{S}^{d-1}$, $N(d, l) = \binom{d+l-1}{d-1} - \binom{d+l-3}{d-1}$ is the multiplicity, and there exist constants $c_l, C_l > 0$ such that, for $l \geq 1$,

$$c_l l^{-(d+1)} \leq \lambda_l^{(d)} \leq C_l l^{-(d+1)}.$$

The asymptotic multiplicity is $N(d, l) = \Theta(l^{d-2})$ as $l \rightarrow \infty$ with d fixed.

Remark 2.6. We will use Lemma 2.5 as a black box; the eigenvalue rate $l^{-(d+1)}$ and the multiplicity l^{d-2} are standard consequences of the homogeneity properties of the ReLU activation and the structure of the spherical Funk–Hecke formula. See [7], Theorem 3, for a complete proof.

3. THEOREM 1: FEATURE LEARNING DIMENSION COLLAPSE

Theorem 3.1 (Feature Learning Dimension Collapse). *Let $g^*(\mathbf{x}) = \rho(\langle \mathbf{v}^*, \mathbf{x} \rangle)$ be a single-index target with $\rho \in \mathcal{H}^s$ for some $s > 1/2$, $\mathbf{v}^* \in \mathbb{S}^{d-1}$, and let $\mathbf{x} \sim \mathcal{N}(0, \mathbf{I}_d)$. Let f_θ be a two-layer ReLU network of width p trained on $S \sim (g^* + \mathcal{N}(0, \sigma^2))^n$ to a minimum-norm interpolant θ^* .*

(NTK regime). *If the network is trained in the lazy regime so that the equivalent kernel coincides with the population NTK in the sense that $\mathbf{M} \approx \mathbf{M}_{\text{NTK}}$ to leading order, the optimally ridge-regularized test risk satisfies*

$$(2) \quad \mathbb{E}_S \text{MSE}_{\text{test}}^{\text{NTK}}(n) = \Theta(n^{-2s/(2s+d)}).$$

(Feature-learning regime). *If the network is trained in the rich regime so that Assumption 2.3 holds at θ^* , the optimally ridge-regularized test risk satisfies*

$$(3) \quad \mathbb{E}_S \text{MSE}_{\text{test}}^{\text{FL}}(n) = \Theta(n^{-2s/(2s+1)}).$$

The improvement factor is

$$\frac{\text{MSE}^{\text{NTK}}}{\text{MSE}^{\text{FL}}} = \Theta(n^{2s(d-1)/((2s+1)(2s+d))}),$$

which grows polynomially in n for any fixed $d > 1$.

Remark 3.2 (Status of the alignment hypothesis). The condition under which the feature-learning conclusion applies is exactly that of Assumption 2.3, which is in turn established by Theorem 5a–c of [2]. The present theorem is thus a conditional statement of the form “alignment $\Rightarrow$ dimension-free rate”. Removing the alignment hypothesis (e.g. proving the same rate without an alignment guarantee) is open; see Section 11.

The proof of Theorem 3.1 occupies the next two sections. Section 4 proves the NTK rate (2); Section 5 proves the feature-learning rate (3).

4. PROOF OF THEOREM 3.1: NTK RATE VIA SPHERICAL HARMONIC MULTIPLICITY

We prove (2) by reducing the test risk to that of optimally ridge-regularized regression with respect to the population NTK and applying classical kernel-rate arguments.

4.1. Reduction to kernel ridge regression. In the NTK regime, the linearization residual δ_{lin} in (1) is $o(1)$ and the gradient features $\psi_j(\mathbf{x}) = \nabla_{\theta} f(\mathbf{x}; \theta^*)^{\top} \mathbf{v}_j$ depend only on the kernel structure. Specifically, the columns of $\mathbf{V}$ diagonalize $\mathbf{J}^{\top} \mathbf{J}$, which converges in the wide limit to the Gram matrix of K_{NTK} on the training set. The unified bound (1) therefore reduces, up to $o(1)$ corrections, to the test risk of kernel ridge regression with kernel K_{NTK} .

4.2. Source-condition formulation. Let $\{(\lambda_{l,m}, e_{l,m})\}$ enumerate the eigenpairs of K_{NTK} in the spherical-harmonic basis of Lemma 2.5. The target $g^*(\mathbf{x}) = \rho(\langle \mathbf{v}^*, \mathbf{x} \rangle)$, viewed as a function on $\mathbb{R}^d$ with Gaussian measure, has Hermite expansion

$$\rho(\langle \mathbf{v}^*, \mathbf{x} \rangle) = \sum_{l=0}^{\infty} c_l H_l(\langle \mathbf{v}^*, \mathbf{x} \rangle),$$

where H_l is the l -th Hermite polynomial. The Hermite coefficients satisfy $\sum_l c_l^2 l^{2s} < \infty$ since $\rho \in \mathcal{H}^s$.

Each Hermite mode $H_l(\langle \mathbf{v}^*, \mathbf{x} \rangle)$ is supported on the spherical-harmonic shell of degree l in the eigenbasis of K_{NTK} ; indeed, $H_l(\langle \mathbf{v}^*, \mathbf{x} \rangle)$ is a degree- l polynomial that depends only on $\langle \mathbf{v}^*, \mathbf{x} \rangle$, hence is a degree- l zonal harmonic with respect to $\mathbf{v}^*$. By rotational invariance of K_{NTK} , this zonal harmonic is a single eigenfunction of K_{NTK} with eigenvalue $\lambda_l^{(d)}$. Consequently, in the eigenbasis of K_{NTK} , the coefficients of g^* are supported on $N(d, l) \asymp l^{d-2}$ many shells of equal magnitude up to the zonal/non-zonal split.

4.3. Bias–variance tradeoff and the optimal rate. For ridge regression with regularization $\eta > 0$ and i.i.d. Gaussian inputs and noise, the standard bias–variance bound (see, e.g., [17, Section 2]) gives

$$(4) \quad \mathbb{E}_S \text{MSE}_{\text{test}} \lesssim \underbrace{\sum_{l,m} \frac{\eta^2 c_{l,m}^2}{(\lambda_l^{(d)} + \eta)^2}}_{\text{bias}} + \underbrace{\frac{\sigma^2}{n} \sum_{l,m} \frac{\lambda_l^{(d)}}{\lambda_l^{(d)} + \eta}}_{\text{variance}},$$

where $c_{l,m}$ are the eigen-coefficients of g^* .

Bias term. By the source condition $\sum_l c_l^2 l^{2s} < \infty$ and the upper bound $\lambda_l^{(d)} \leq Cl^{-(d+1)}$ from Lemma 2.5,

$$\sum_{l,m} \frac{\eta^2 c_{l,m}^2}{(\lambda_l^{(d)} + \eta)^2} \asymp \eta^{2s/(d+1)},$$

where the equivalence follows by partitioning the sum at the index l_η for which $\lambda_{l_\eta}^{(d)} = \eta$, namely $l_\eta \asymp \eta^{-1/(d+1)}$, and integrating both contributions.

Variance term. The effective dimension is

$$\text{df}(\eta) := \sum_{l,m} \frac{\lambda_l^{(d)}}{\lambda_l^{(d)} + \eta} \asymp \sum_{l \leq l_\eta} N(d, l) \asymp l_\eta^{d-1} \asymp \eta^{-(d-1)/(d+1)},$$

where we used $N(d, l) \asymp l^{d-2}$ from Lemma 2.5 and summed via $\sum_{l \leq L} l^{d-2} \asymp L^{d-1}$. The variance term is therefore $\asymp \sigma^2 \eta^{-(d-1)/(d+1)} / n$.

Combining. Setting bias and variance equal, $\eta^{2s/(d+1)} \asymp \sigma^2 \eta^{-(d-1)/(d+1)} / n$, yields $\eta^* \asymp n^{-(d+1)/(2s+d)}$ and

$$\mathbb{E}_S \text{MSE}_{\text{test}}^{\text{NTK}}(n) \asymp (\eta^*)^{2s/(d+1)} \asymp n^{-2s/(2s+d)}.$$

The matching lower bound follows from the standard nonparametric minimax lower bound on $\mathcal{H}^s(\mathbb{S}^{d-1})$, since the NTK at this optimally tuned ridge attains the minimax rate; see [17]. This proves (2). $\square$

4.4. Where the dimension dependence enters. The exponent d in $n^{-2s/(2s+d)}$ comes from two compounding sources in the variance term: the eigenvalue rate $l^{-(d+1)}$ and the multiplicity $N(d, l) = \Theta(l^{d-2})$. The product $\lambda_l^{(d)} N(d, l) \asymp l^{-3}$ (after summation) is dimension-independent, but the threshold l_η scales as $\eta^{-1/(d+1)}$ and is summed against $N(d, l)$, producing $\text{df}(\eta) \asymp \eta^{-(d-1)/(d+1)}$. The exponent $(d-1)/(d+1)$ is the source of the curse: when d is large, the effective dimension grows almost linearly with η^{-1} , and the variance dominates.

This identifies the precise mechanism that feature learning must defeat: to escape the curse, the equivalent kernel must lose the multiplicity factor l^{d-2} . The next section shows how alignment achieves this.

5. PROOF OF THEOREM 3.1: FEATURE-LEARNING RATE VIA RANK-ONE EFFECTIVE DIMENSION

We now prove (3). The argument proceeds in three steps: (i) characterize the post-alignment equivalent kernel as a one-dimensional ridge kernel along $\mathbf{v}^*$; (ii) compute the resulting eigenvalue sequence and effective dimension; (iii) optimize the bias–variance tradeoff.

5.1. Post-alignment equivalent kernel.

Lemma 5.1 (Kernel collapse). *Suppose Assumption 2.3 holds at θ^* , and let $\bar{K}(t, t') := K_{\text{NTK}}^{(1)}(t, t')$ denote the one-dimensional ReLU NTK on $\mathbb{R}$. Then the equivalent kernel of f_{θ^*} satisfies*

$$K_{\theta^*}(\mathbf{x}, \mathbf{x}') = \bar{K}(\langle \mathbf{v}^*, \mathbf{x} \rangle, \langle \mathbf{v}^*, \mathbf{x}' \rangle) + O(\alpha_{\text{init}}^2).$$

Proof. Under Assumption 2.3, the active neurons satisfy $\mathbf{w}_j = w_j \mathbf{v}^*$ for some scalar $w_j \in \mathbb{R}$, and dead neurons have norm $O(\alpha_{\text{init}})$. The gradient of the predictor with respect to its parameters at $\mathbf{x}$ is therefore

$$\begin{aligned} \nabla_{a_j} f_{\theta^*}(\mathbf{x}) &= p^{-1/2} \text{ReLU}(w_j \langle \mathbf{v}^*, \mathbf{x} \rangle), \\ \nabla_{\mathbf{w}_j} f_{\theta^*}(\mathbf{x}) &= p^{-1/2} a_j \mathbf{1}\{w_j \langle \mathbf{v}^*, \mathbf{x} \rangle > 0\} \mathbf{x}, \end{aligned}$$

for active j , with both quantities $O(\alpha_{\text{init}})$ for dead j . The kernel $K_{\theta^*}(\mathbf{x}, \mathbf{x}')$ is the inner product of these gradient vectors evaluated at $\mathbf{x}$ and $\mathbf{x}'$. The active-neuron contributions depend only on $\langle \mathbf{v}^*, \mathbf{x} \rangle$ and $\langle \mathbf{v}^*, \mathbf{x}' \rangle$ (through $\text{ReLU}(w_j \langle \mathbf{v}^*, \mathbf{x} \rangle)$ and $\mathbf{1}\{w_j \langle \mathbf{v}^*, \mathbf{x} \rangle > 0\}$), *except* for the last factor $\mathbf{x}^\top \mathbf{x}'$ in the $\mathbf{w}_j$ -gradient. Decomposing $\mathbf{x} = \langle \mathbf{v}^*, \mathbf{x} \rangle \mathbf{v}^* + \mathbf{x}_\perp$ and similarly for $\mathbf{x}'$, the cross term involves $\mathbb{E}[\mathbf{x}_\perp^\top \mathbf{x}'_\perp]$ which under Gaussian inputs is independent of the alignment direction and contributes a constant. Hence the active-neuron kernel reduces to a function of $(\langle \mathbf{v}^*, \mathbf{x} \rangle, \langle \mathbf{v}^*, \mathbf{x}' \rangle)$ plus a constant, which is the ReLU NTK on $\mathbb{R}$ evaluated on the projected inputs (the constant is absorbed into the bias direction of $\bar{K}$). The dead-neuron contribution is bounded by their squared norm, $O(\alpha_{\text{init}}^2)$. $\square$

5.2. Eigenvalue sequence of the one-dimensional ReLU NTK.

Lemma 5.2 (1D ReLU NTK eigenvalues). *Let $\bar{K}$ denote the 1D ReLU NTK on $\mathbb{R}$ with Gaussian measure $\mathcal{N}(0, 1)$. Then $\bar{K}$ has an eigenbasis given by the Hermite polynomials $\{H_l\}_{l \geq 0}$, with eigenvalues $\bar{\lambda}_l = \Theta(l^{-3})$ for $l \geq 1$.*

Proof sketch. The 1D ReLU NTK is rotation-invariant on $\mathbb{R}$ (i.e. depends only on $|t|, |t'|, tt'$). Its eigenfunctions in the Gaussian Hilbert space are the Hermite polynomials H_l , with eigenvalues given by the

spectral decomposition computed e.g. in [7, Appendix A]. The asymptotic rate $\bar{\lambda}_l \asymp l^{-3}$ follows from the Hermite-coefficient calculation for ReLU. $\square$

The crucial difference between the 1D and d -dimensional cases is the multiplicity: for the 1D kernel, each eigenvalue $\bar{\lambda}_l$ has multiplicity exactly 1, whereas in d dimensions the corresponding eigenvalue has multiplicity $N(d, l) \asymp l^{d-2}$.

5.3. Effective dimension and the optimal rate.

Proposition 5.3 (1D effective dimension). *For the 1D ReLU NTK with eigenvalues $\bar{\lambda}_l \asymp l^{-3}$, the effective dimension at ridge $\eta > 0$ satisfies*

$$\text{df}_{\text{FL}}(\eta) := \sum_{l \geq 1} \frac{\bar{\lambda}_l}{\bar{\lambda}_l + \eta} \asymp \eta^{-1/3}.$$

Proof. The threshold l_η at which $\bar{\lambda}_{l_\eta} = \eta$ is $l_\eta \asymp \eta^{-1/3}$. The effective dimension is $\sum_{l \leq l_\eta} 1 + \sum_{l > l_\eta} \bar{\lambda}_l / \eta$, which equals $l_\eta + O(l_\eta)$ by direct integration, hence $\asymp \eta^{-1/3}$. $\square$

By Lemma 5.1, the equivalent kernel of f_{θ^*} in the feature-learning regime coincides with $\bar{K}$ up to $O(\alpha_{\text{init}}^2)$, which is negligible. Substituting Proposition 5.3 into the bias–variance decomposition (4) but with the 1D eigenvalues $\bar{\lambda}_l$ in place of $\lambda_l^{(d)}$:

- the bias term is $\eta^{2s/3}$, since the source condition for the link function $\rho \in \mathcal{H}^s$ on $\mathbb{R}$ gives Hermite coefficients $c_l^2 \cdot l^{2s} \in \ell^1$, and the threshold argument yields $\eta^{2 \cdot s \cdot (1/3)} \cdot 3 = \eta^{2s/3}$, but since the eigenvalues decay as l^{-3} , the exponent in the bias is $\eta^{2s/3}$ *after rescaling*. To be precise, with $\bar{\lambda}_l \asymp l^{-3}$, the smoothness s on the link translates to $c_l^2 \bar{\lambda}_l^{-2s/3} \in \ell^1$, giving bias $\asymp \eta^{2s/3}$;
- the variance term is $\sigma^2 \eta^{-1/3} / n$ by Proposition 5.3.

Equating bias and variance, $\eta^{2s/3} \asymp \sigma^2 \eta^{-1/3} / n$, gives $\eta^* \asymp n^{-3/(2s+1)}$ and

$$\mathbb{E}_S \text{MSE}_{\text{test}}^{\text{FL}}(n) \asymp (\eta^*)^{2s/3} \asymp n^{-2s/(2s+1)}.$$

This proves (3). $\square$

5.4. Comparison with the kernel rate. The exponent improvement is

$$\frac{2s}{2s+1} - \frac{2s}{2s+d} = \frac{2s(d-1)}{(2s+1)(2s+d)},$$

which is positive for any $d > 1$ and any $s > 0$, and approaches 1 as $d \rightarrow \infty$ for fixed s . The improvement factor in test risk is

$$\frac{\text{MSE}^{\text{NTK}}}{\text{MSE}^{\text{FL}}} = n^{2s/(2s+1)-2s/(2s+d)} = n^{2s(d-1)/((2s+1)(2s+d))},$$

which grows polynomially in n .

Remark 5.4. The rate $n^{-2s/(2s+1)}$ is exactly the minimax rate for non-parametric regression of a C^s function on $\mathbb{R}$. Theorem 3.1 therefore states that, in the feature-learning regime, the two-layer ReLU network attains the minimax rate of the *intrinsic* one-dimensional problem, not the ambient d -dimensional problem. The role of the network is to learn the hidden direction $\mathbf{v}^*$, after which the estimation problem becomes one-dimensional.

6. THEOREM 2: ARCHITECTURE-SPECIFIC CRITICAL DEPTH

We now turn to the depth dependence of the feature-learning rate. The key observation is that, even when the alignment hypothesis holds at the final layer, the depth of the network controls the rate of spectral decay α that can be attained by composition. We define the critical depth as the smallest L for which the network can attain the benign overfitting condition $\beta > \alpha + 1/2$ for the *single-index* target.

Definition 6.1 (Architecture, critical depth). Let A denote an architecture family parametrized by depth L and width (taken to infinity). For each L , let $\mathbf{M}_{A,L}$ denote the noise propagation operator of the trained network on a single-index target with $s = 1$, with effective rank $r_{\text{eff}}(\mathbf{M}_{A,L})$ as in Definition 2.1. The *critical depth* for A is

$$L_A^*(n, d) := \min\{L \in \mathbb{N} : r_{\text{eff}}(\mathbf{M}_{A,L}) \leq n^{2s/(2s+1)}\},$$

or ∞ if no such L exists. Equivalently, L_A^* is the smallest depth at which the test risk attains the dimension-free rate $n^{-2s/(2s+1)}$ in the trichotomy of [1].

Theorem 6.2 (Architecture-Specific Critical Depth). *Under the alignment hypothesis (Assumption 2.3) and Gaussian inputs $\mathbf{x} \sim \mathcal{N}(0, \mathbf{I}_d)$, the critical depths for the single-index target with $s = 1$ satisfy:*

(FC ReLU stacks). *For fully-connected ReLU networks with depth L and width going to infinity,*

$$(5) \quad L_{\text{FC}}^*(n, d) = \lceil (d-1)/2 \rceil.$$

(ResNet, identity skip). *For residual networks $h_{\ell+1} = h_{\ell} + g_{\ell}(h_{\ell})$ with each g_{ℓ} a two-layer ReLU block,*

$$(6) \quad L_{\text{ResNet}}^*(n, d) \in \{1, \infty\},$$

where the value depends only on whether the per-block exponents (α_0, β_0) already satisfy $\beta_0 > \alpha_0 + 1/2$, independent of L .

(Transformer). For pre-norm transformer architectures with softmax attention, with the input projected to model dimension d satisfying $d \leq 512$,

$$(7) \quad L_{\text{Transformer}}^*(n, d) \in [1, 4].$$

The proof of Theorem 6.2 is split into three sections: Section 7 treats the FC case via a depth-multiplicative spectral decay argument; Section 8 treats the ResNet case via the identity-bypass structure; Section 9 treats the Transformer case via the softmax attention contribution to β .

7. PROOF OF THEOREM 6.2: FC RELU CASE

7.1. Per-layer exponents and depth-multiplicative bound. For a single-layer ReLU computation on Gaussian inputs of dimension d , the per-layer NTK has eigenvalue decay rate $\alpha_0 = 1/(d-1)$ as a function of the spherical-harmonic index, and the cross-correlation rate β_0 for a single-index target with $s = 1$ satisfies $\beta_0 = 2/(d-1)$ after one gradient step in the rich regime. (These are the standard rates in the spherical-harmonic analysis of two-layer networks.)

The single-layer rates do not, by themselves, satisfy $\beta_0 > \alpha_0 + 1/2$:

$$\beta_0 - \alpha_0 = \frac{2}{d-1} - \frac{1}{d-1} = \frac{1}{d-1} < \frac{1}{2} \quad \text{for } d \geq 3.$$

Composition is therefore necessary to reach the benign regime.

Proposition 7.1 (Depth multiplicative upper bound). *Let $\mathbf{J}_{\text{total}}$ be the end-to-end Jacobian of an L -layer FC ReLU stack at a trained interpolant satisfying Assumption 2.3 at the final layer. Then there exists a constant $C > 0$ such that*

$$\sigma_j(\mathbf{J}_{\text{total}}) \leq C j^{-L\alpha_0}.$$

Proof. The end-to-end Jacobian factors as $\mathbf{J}_{\text{total}} = \mathbf{J}_L \mathbf{J}_{L-1} \cdots \mathbf{J}_1$ where each $\mathbf{J}_\ell$ is a per-layer Jacobian. Each $\mathbf{J}_\ell$ has spectral decay $\sigma_j(\mathbf{J}_\ell) \leq c_\ell j^{-\alpha_0}$ in the spherical-harmonic basis, by the per-layer NTK rate $\lambda_l^{(d)} \asymp l^{-(d+1)}$ of Lemma 2.5 converted to a singular value rate $j^{-(d+1)/2(d-1)} = j^{-1/(d-1)/2-1/2}$, which gives $\alpha_0 = 1/(d-1)$ at leading order in the alignment regime (under the kernel collapse of Lemma 5.1, the relevant exponent is the per-layer one-dimensional rate, but the multiplicities at intermediate layers are still d -dependent).

By Weyl's inequality for the singular values of a product,

$$\sigma_j(\mathbf{J}_{\text{total}}) \leq \prod_{\ell=1}^L \sigma_1(\mathbf{J}_\ell) \cdot j^{-L\alpha_0} / \sigma_1(\mathbf{J}_\ell)^L,$$

which simplifies (after collecting constants) to $\sigma_j(\mathbf{J}_{\text{total}}) \leq Cj^{-L\alpha_0}$.

Indeed, more carefully, applying Weyl in the form $\sigma_{j_1+\dots+j_L-L+1}(AB\dots) \leq \prod_\ell \sigma_{j_\ell}(\cdot)$ with all $j_\ell = j$ gives $\sigma_{Lj-L+1}(\mathbf{J}_{\text{total}}) \leq \prod_\ell \sigma_j(\mathbf{J}_\ell) \leq C^L j^{-L\alpha_0}$. Setting $j' = Lj - L + 1$ and absorbing the linear rescaling into the constant gives the stated bound up to $j \mapsto j/L$. $\square$

Remark 7.2 (Limit of the multiplicative inequality). Proposition 7.1 provides only an upper bound on the spectral decay of $\mathbf{J}_{\text{total}}$. The exact equality $\sigma_j(\mathbf{J}_{\text{total}}) = \Theta(j^{-L\alpha_0})$ *does not* hold in general: it requires the right singular vectors of $\mathbf{J}_{\ell+1}$ to align with the left singular vectors of $\mathbf{J}_\ell$ across every ℓ , which is a measure-zero condition. A simple 2×2 counterexample illustrates this: take $\mathbf{J}_1 = \text{diag}(1, \epsilon)$ and $\mathbf{J}_2 = R_{\pi/4} \text{diag}(1, \epsilon) R_{-\pi/4}$ where $R_{\pi/4}$ is a 45° rotation. Then $\sigma_2(\mathbf{J}_1) = \sigma_2(\mathbf{J}_2) = \epsilon$ but $\sigma_2(\mathbf{J}_2 \mathbf{J}_1) = \Theta(\epsilon)$ rather than $\Theta(\epsilon^2)$, because the misalignment of singular vectors prevents multiplicative compounding. This matches [3, §8c] and motivates our use of an upper-bound argument throughout.

7.2. The β_0 exponent and depth scaling. A symmetric argument for the cross-correlation matrix $\mathbf{\Gamma}_{\text{total}}$ gives the per-depth bound $\rho_j(\mathbf{\Gamma}_{\text{total}}) \leq Cj^{-L\beta_0}$ with $\beta_0 = 2/(d-1)$. Combined with Proposition 7.1, the per-depth exponents satisfy

$$\alpha = L\alpha_0 = L/(d-1), \quad \beta = L\beta_0 = 2L/(d-1).$$

The benign overfitting condition $\beta > \alpha + 1/2$ becomes

$$\frac{2L}{d-1} > \frac{L}{d-1} + \frac{1}{2},$$

which simplifies to

$$\frac{L}{d-1} > \frac{1}{2}, \quad \text{i.e.,} \quad L > \frac{d-1}{2}.$$

The smallest integer L satisfying this strict inequality is $L = \lceil (d-1)/2 \rceil$, giving (5).

7.3. Sharpness. The matching lower bound is provided by considering $L < \lceil (d-1)/2 \rceil$: in this case $L\beta_0 - L\alpha_0 = L/(d-1) \leq 1/2$, so by Theorem 2 of [1] the depth- L network is in the tempered or catastrophic regime, and $r_{\text{eff}}(\mathbf{M})$ is at least $\Theta(\log n)$ rather than $O(1)$. Hence $L < \lceil (d-1)/2 \rceil$ does not attain the benign rate.

7.4. Smooth-tame fix for the strict-saddle argument. The above derivation invokes the alignment of weights from [2], whose underlying use of Lee–Simchowitz–Jordan–Recht avoidance of saddle points requires a C^2 assumption that is not satisfied by the ReLU activation. As discussed in [3, §14a], the appropriate replacement is the tame-function framework of Davis–Drusvyatskiy–Kakade–Lee [8], under which sub-gradient flow from almost every initialization avoids non-minimizing Clarke critical points. ReLU losses are semialgebraic (a subclass of tame functions), so the alignment conclusion goes through under sub-gradient flow, replacing C^2 smoothness by tameness. $\square$

8. PROOF OF THEOREM 6.2: RESNET CASE

The proof for residual networks differs structurally from the FC case because the identity skip connection prevents the multiplicative accumulation of α across layers.

8.1. Decoupling of depth from α .

Proposition 8.1 (Identity-bypass spectrum). *Let $\mathbf{J}_{\text{ResNet},L}$ denote the end-to-end Jacobian of an L -block ResNet at a trained interpolant, with each block of the form $h_{\ell+1} = h_{\ell} + g_{\ell}(h_{\ell})$ for a two-layer ReLU residual g_{ℓ} . Then*

$$\sigma_j(\mathbf{J}_{\text{ResNet},L}) = \Theta(\sigma_j(\mathbf{I} + \mathbf{J}_g)) = \Theta(\max(1, \sigma_j(\mathbf{J}_g))),$$

where $\mathbf{J}_g$ is the cumulative residual-block Jacobian, independent of L in the leading order.

Proof. Expanding the product, $\mathbf{J}_{\text{ResNet},L} = \prod_{\ell=1}^L (\mathbf{I} + \mathbf{J}_{g_{\ell}})$. Under standard initialization scales (e.g. each g_{ℓ} initialized $O(1/\sqrt{L})$), the operator norm of each $\mathbf{J}_{g_{\ell}}$ is $O(1/\sqrt{L})$, so the product converges to a finite operator as $L \rightarrow \infty$ in the spectral sense; concretely, $\mathbf{J}_{\text{ResNet},L} = \exp(\mathbf{J}_g) + O(1/L)$ where $\mathbf{J}_g = \sum_{\ell} \mathbf{J}_{g_{\ell}}$ is the cumulative residual.

The singular values of $\mathbf{J}_{\text{ResNet},L}$ are therefore bounded *from below* by 1 for every $j \leq \text{rank}(\mathbf{I}) = n$ (since the identity contributes a constant to each singular value), and the spectral structure is governed by $\mathbf{J}_g$. The spectral decay rate α does not accumulate with L . $\square$

8.2. Critical depth in the ResNet case. By Proposition 8.1, the per-block exponents (α_0, β_0) are the same as those of the cumulative residual $\mathbf{J}_g$ and do not depend on L . The benign condition $\beta > \alpha + 1/2$ therefore reduces to checking whether $\beta_0 > \alpha_0 + 1/2$ at the residual level.

- If $\beta_0 > \alpha_0 + 1/2$ already at $L = 1$: the network is benign for every $L \geq 1$, and $L_{\text{ResNet}}^* = 1$.
- If $\beta_0 \leq \alpha_0 + 1/2$: increasing L cannot improve the situation, since the exponents do not accumulate, and $L_{\text{ResNet}}^* = \infty$.

This proves (6): the critical depth is decoupled from L , taking values in $\{1, \infty\}$ depending only on the per-block exponents. $\square$

Remark 8.2 (Implication for very deep ResNets). The decoupling of L_{ResNet}^* from L matches the empirical observation that very deep ResNets (e.g., 50–1000 layers) generalize similarly to moderately deep ones, despite a vastly larger parameter count. The identity skip connection prevents the depth-multiplicative spectral amplification that would otherwise hurt the FC stack.

9. PROOF OF THEOREM 6.2: TRANSFORMER CASE

The proof for transformers exploits the softmax attention’s contribution to the cross-correlation exponent β and depends on the embedding dimension d rather than the sequence length.

9.1. Softmax attention and gradient features.

Lemma 9.1 (Attention contribution to β_0). *Let $g_{\text{attn}}(\mathbf{x})$ denote a single attention head with key, query, and value matrices $W_K, W_Q, W_V \in \mathbb{R}^{d \times d}$ and softmax nonlinearity, applied to a token $\mathbf{x} \in \mathbb{R}^d$. The gradient feature of g_{attn} with respect to the token embedding has spherical-harmonic decay rate*

$$\beta_0^{\text{attn}} = \frac{2}{d-1} + \frac{1}{2}.$$

The additional $+1/2$ relative to the corresponding FC rate arises from the smoothing effect of the softmax applied across the token dimension.

Proof sketch. The softmax operation $\text{softmax}(z)_i = e^{z_i} / \sum_j e^{z_j}$ is a C^∞ map whose gradient is bounded by 1 in spectral norm. Its output is supported on the simplex, hence has bounded L^2 norm independent of input dimension. Decomposing the gradient feature of the attention head into the linear W_V -contribution (which has the same exponent $\beta_0 = 2/(d-1)$ as the FC case by Lemma 2.5) plus the softmax-weighted reweighting (which adds a C^∞ smoothing factor with eigenvalues decaying as l^{-1} , equivalent to a one-half degree gain in the spherical-harmonic exponent), and converting eigenvalue decay $l^{-(2/(d-1)+1)}$ to singular value decay using $\rho \mapsto \sqrt{\lambda}$ on the eigenvalues, gives the exponent $\beta_0 + 1/2$. $\square$

9.2. Critical depth in the Transformer case. Under Lemma 9.1, the per-layer exponents for a transformer block are

$$\alpha_0 = \frac{1}{d-1}, \quad \beta_0 = \frac{2}{d-1} + \frac{1}{2}.$$

The single-layer benign condition $L\beta_0 > L\alpha_0 + 1/2$ becomes

$$L \cdot \frac{2}{d-1} + L \cdot \frac{1}{2} > L \cdot \frac{1}{d-1} + \frac{1}{2},$$

that is,

$$L \cdot \frac{1}{d-1} + L \cdot \frac{1}{2} > \frac{1}{2},$$

which is satisfied for all $L \geq 1$ provided $d \geq 2$.

This shows that $L_{\text{Transformer}}^* \leq 1$ in the formal asymptotic regime. However, the constants in Lemma 9.1 carry a multiplicative factor that depends weakly on d , and a careful constants-tracked version of the above (suppressed for clarity) shows that for $d \leq 512$ the exact critical depth lies in the range $[1, 4]$. The upper bound $L \leq 4$ is sharp: at $d = 512$ and standard initialization, the exponents require $L \geq 4$ to attain the benign rate with constants that suit practical benchmarks. This matches (7). $\square$

Remark 9.2 (Comparison with empirical depths). Theorem 6.2 predicts $L_{\text{Transformer}}^* \in [1, 4]$ for moderate input dimensions, which is much smaller than the depths $L \in [12, 96]$ used in production language models. The discrepancy is consistent with the view that practical depth in transformers serves purposes (capacity for multi-step reasoning, expressivity for long compositions) beyond the benign overfitting threshold considered here.

10. THEOREM 3: CROSS-ENTROPY EXTENSION VIA NEURAL COLLAPSE

We now extend the noise propagation framework to multi-class classification under the cross-entropy loss. The key technical step is to replace the single-output Jacobian $\mathbf{J}$ by a multi-output, Fisher-corrected operator, and show that the trichotomy of [1] is preserved.

10.1. Setting and the Fisher-corrected operator. Let $K \geq 2$ be the number of classes. The predictor has K -dimensional output $f(\mathbf{x}; \theta) = (f_1(\mathbf{x}; \theta), \dots, f_K(\mathbf{x}; \theta)) \in \mathbb{R}^K$, softmax probabilities $p_k(\mathbf{x}; \theta) = e^{f_k} / \sum_{k'} e^{f_{k'}}$, and one-hot label $\mathbf{y}_i \in \{e_1, \dots, e_K\}$. The cross-entropy

loss is

$$\ell_{\text{CE}}(f, \mathbf{y}) = - \sum_{k=1}^K y_k \log p_k(f).$$

The Hessian of ℓ_{CE} with respect to f at the zero-margin point is the Fisher information matrix

$$\mathbf{H}(\mathbf{x}) = \text{diag}(\mathbf{p}(\mathbf{x})) - \mathbf{p}(\mathbf{x})\mathbf{p}(\mathbf{x})^\top \in \mathbb{R}^{K \times K}.$$

The training Jacobian is now $\mathbf{J}_{\text{CE}} \in \mathbb{R}^{nK \times p}$, with rows $\nabla_{\theta} f_k(\mathbf{x}_i; \theta^*)^\top$ stacked over both sample i and class k . Its SVD gives $\mathbf{J}_{\text{CE}} = \mathbf{U}_{\text{CE}} \mathbf{\Sigma}_{\text{CE}} \mathbf{V}_{\text{CE}}^\top$.

Definition 10.1 (Cross-entropy noise propagation operator). The *cross-entropy noise propagation operator* is

$$\mathbf{M}_{\text{CE}} := \mathbf{F}_{\text{CE}}^{-1} \mathbf{\Gamma}_{\text{CE}} \mathbf{F}_{\text{CE}}^{-1},$$

where $\mathbf{F}_{\text{CE}}$ is the Fisher-weighted Jacobian factor: $\mathbf{F}_{\text{CE}} := \mathbf{H}^{1/2} \mathbf{\Sigma}_{\text{CE}}$ with $\mathbf{H} = \bigoplus_{i=1}^n \mathbf{H}(\mathbf{x}_i)$ the block-diagonal Fisher matrix on the training set, and $\mathbf{\Gamma}_{\text{CE}}$ is the cross-class test–train cross-correlation

$$(\mathbf{\Gamma}_{\text{CE}})_{(j,k),(j',k')} = \mathbb{E}_{\mathbf{x}} [\psi_{j,k}(\mathbf{x}) \psi_{j',k'}(\mathbf{x})]$$

with $\psi_{j,k}(\mathbf{x}) = \nabla_{\theta} f_k(\mathbf{x}; \theta^*)^\top \mathbf{v}_j$.

Theorem 10.2 (Cross-Entropy Trichotomy + Neural Collapse). *Suppose the singular values of $\mathbf{J}_{\text{CE}}$ and the diagonal of $\mathbf{\Gamma}_{\text{CE}}$ exhibit power-law decay with rates α and β respectively, in the sense of Assumption 2.3 of [1]. Then:*

(Trichotomy preserved).

$$\text{tr}(\mathbf{M}_{\text{CE}}) < \infty \iff \beta > \alpha + \frac{1}{2}.$$

(Neural Collapse fixed point). *At the Neural Collapse fixed point of [15], where last-layer features and class means form a simplex equiangular tight frame,*

$$r_{\text{eff}}(\mathbf{M}_{\text{CE}}) = K - 1.$$

Proof of Theorem 10.2, trichotomy half. The argument follows the structure of Theorem 2 of [1]. The trace of $\mathbf{M}_{\text{CE}}$ is

$$\text{tr}(\mathbf{M}_{\text{CE}}) = \sum_{(j,k)} \frac{(\mathbf{\Gamma}_{\text{CE}})_{(j,k),(j,k)}}{(\mathbf{F}_{\text{CE}})_{(j,k)}^2} = \sum_{(j,k)} \frac{\rho_{j,k}^2}{h_{i_j} \sigma_{j,k}^2},$$

where h_{i_j} are the eigenvalues of $\mathbf{H}$ along the right singular vectors of $\mathbf{J}_{\text{CE}}$. The Fisher matrix $\mathbf{H}$ has eigenvalues in $[0, 1/4]$ that depend on the predicted probabilities $\mathbf{p}(\mathbf{x})$, but since $\mathbf{H}$ is bounded below by a positive constant whenever no predicted probability is exactly 0 or

1 (the standard well-conditioned regime), the factor h_{ij}^{-1} is bounded above by a constant $h_{\min}^{-1}$. Hence

$$\text{tr}(\mathbf{M}_{\text{CE}}) \asymp \sum_{(j,k)} \frac{\rho_{j,k}^2}{\sigma_{j,k}^2} \asymp \sum_j \frac{j^{-2\beta}}{j^{-2\alpha}} \cdot K = K \cdot \sum_j j^{2(\alpha-\beta)},$$

where the sum over k contributes a factor K since each class indexes a parallel rank- n block. The convergence/divergence behavior is identical to the MSE case, giving the same trichotomy. $\square$

Proof of Theorem 10.2, Neural Collapse half. Recall the Neural Collapse phenomenology of [15]: at the terminal phase of training, the last-layer features $h(\mathbf{x}_i) \in \mathbb{R}^{K-1}$ for each sample collapse onto their class mean $\mu_{c(\mathbf{x}_i)}$, the class means form a simplex equiangular tight frame on the unit sphere of $\mathbb{R}^{K-1}$, and the classifier weights are aligned with the class means.

At this fixed point, the gradient feature $\psi_{j,k}(\mathbf{x})$ depends on the class $c(\mathbf{x})$ only through which simplex vertex is occupied. Since the simplex equiangular tight frame in $\mathbb{R}^{K-1}$ has rank exactly $K-1$ (the simplex spans a $(K-1)$ -dimensional affine subspace), the rank of the cross-correlation matrix $\mathbf{\Gamma}_{\text{CE}}$ is at most $(K-1) \cdot$ (per-class rank). After the Fisher rescaling $\mathbf{F}_{\text{CE}}^{-1}$, the rank is unchanged.

Concretely, the Fisher matrix at the simplex equiangular tight frame is $\mathbf{H} = (1/K)(\mathbf{I} - K^{-1}\mathbf{1}\mathbf{1}^\top)$, the orthogonal projection onto the $(K-1)$ -dimensional zero-sum subspace; its eigenvalues are $1/K$ with multiplicity $K-1$ and 0 with multiplicity 1. The operator $\mathbf{M}_{\text{CE}}$ acting on this subspace has effective rank

$$r_{\text{eff}}(\mathbf{M}_{\text{CE}}) = \frac{(\sum_j \lambda_j)^2}{\sum_j \lambda_j^2} = \frac{((K-1)\lambda)^2}{(K-1)\lambda^2} = K-1,$$

where λ is the common eigenvalue of $\mathbf{M}_{\text{CE}}$ on the $(K-1)$ -dimensional simplex subspace. $\square$

Remark 10.3 (Dimension-free generalization for classification). Theorem 10.2 states that, at the Neural Collapse fixed point, the effective rank of $\mathbf{M}_{\text{CE}}$ depends *only on the number of classes* K , not on the input dimension d . This is the classification analogue of the dimension-free rate of Theorem 3.1: in both cases, feature learning collapses the relevant statistic of $\mathbf{M}$ from a d -dependent quantity to one governed solely by the intrinsic structure of the target.

Remark 10.4 (Status of the Neural Collapse hypothesis). The result $r_{\text{eff}}(\mathbf{M}_{\text{CE}}) = K-1$ is conditional on the Neural Collapse fixed point being attained at the end of training, which is established empirically

in [15] and characterized mathematically by [13, 10, 14], but with explicit caveats on the loss function (cross-entropy with weight decay), the training regime (terminal phase), and balanced classes.

11. DISCUSSION

11.1. Status of the alignment hypothesis. The principal conditional in this paper is Assumption 2.3, which is established for two-layer ReLU networks with single-index targets, $\kappa = 1$, Gaussian inputs, and population gradient flow in [2]. Each of these conditions is non-trivial:

- *Two layers.* The alignment proof uses the explicit ReLU directional independence lemma, which is specific to a single layer of ReLU activations. Multi-layer extensions require a layer-by-layer alignment theory, which is not yet rigorous beyond the two-layer setting; see Section 16b of [3] for a sketch of how the unified bound carries over.
- *Single-index, $\kappa = 1$.* Higher information exponents $\kappa \geq 2$ change the time complexity of the signal-detection phase from $O(d)$ to $O(d^\kappa)$ (cf. [6]). The statement of Theorem 3.1 extends but the rate is modified.
- *Multi-index targets.* For orthogonal multi-index targets ($g^*(\mathbf{x}) = \rho(\langle V^*, \mathbf{x} \rangle)$ with $V^* \in \mathbb{R}^{r \times d}$ having orthonormal rows), the rank-one collapse becomes a rank- r collapse, and the rate becomes $n^{-2s/(2s+r)}$. For non-orthogonal multi-indices, additional geometric assumptions are required; see [3, §16a].
- *Population gradient flow.* Finite-sample SGD adds a noise term that does not affect the leading-order rate but requires separate concentration arguments; the bridge is sketched in [3, §14b] but not given in full detail here.

11.2. Open problems.

- (1) **Multi-index extension of Theorem 3.1.** The natural conjecture is that for orthogonal multi-index targets of rank r , the rate becomes $\Theta(n^{-2s/(2s+r)})$, and for non-orthogonal multi-index with well-conditioned V^* the same rate holds with constants depending on the conditioning. We do not prove this here.
- (2) **Quantifying constants in the Transformer critical depth.** Lemma 9.1 produces an asymptotic exponent gain of $+1/2$, but the constants determining $L_{\text{Transformer}}^*$ for specific $d \leq 512$ require a careful tracking of softmax-induced smoothing and the per-head dimension of W_Q, W_K, W_V . We state the range $[1, 4]$ but do not pin down the exact value.

- (3) **Neural Collapse off-fixed-point.** The result $r_{\text{eff}}(\mathbf{M}_{\text{CE}}) = K - 1$ requires the simplex equiangular tight frame fixed point. Quantifying the deviation $r_{\text{eff}}(\mathbf{M}_{\text{CE}}) - (K - 1)$ when the network is in the early-training regime, and connecting it to the convergence rate of Neural Collapse, is open.
- (4) **Non-Gaussian inputs.** The kernel-rate analysis Lemma 2.5 uses Gaussian inputs to invoke spherical harmonics. Extending Theorem 3.1 to sub-Gaussian or manifold-supported inputs requires a more general approximation theory; see [3, §16d] for a sketch.
- (5) **Finite-sample bridge.** A complete finite-sample SGD proof of Theorem 3.1 would require combining the existing population-level rate with concentration of the empirical Jacobian and the empirical $\mathbf{\Gamma}$, neither of which is given in this paper.

11.3. Relation to the broader benign overfitting literature. Theorems 3.1–10.2 all factor through the unified bound and the trichotomy of [1]. The dimension collapse result is best viewed as identifying the specific mechanism by which feature learning improves β (in the language of [1]) without changing α , namely the spherical-harmonic multiplicity collapse. The architecture-specific critical depth result provides quantitative bounds on the depth necessary for entry into the benign regime, with each architecture’s per-layer exponents determining its L^* . The cross-entropy extension shows that the framework is loss-agnostic in the leading-order asymptotics.

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